

Pumps & Motors: Compliance, SOP & Optimization Guide

Purpose:

Circulates fluids and drives mechanical systems in HVAC, process cooling, plumbing, and fire suppression operations.

Why It's Ignored:

Assumed to run indefinitely; failures often go unnoticed until production or flow rates drop.

Must-Have Docs:

- Preventive maintenance logs
- Vibration and alignment test reports
- SOPs for shutdown/startup and lubrication
- Motor amp trend charts

Compliance Officials:

OSHA 1910.212, NEC(NFPA 70), Manufacturer Specs

PPE Required:

Gloves, safety goggles, hearing protection, arc-rated gear (for motor panels)

Installation Practices:

- Proper pump/motor alignment
- Secure base mounting with vibration isolation
- Accessible couplings and lubrication points
- Correct rotation and grounding

Must Checks:

- Vibration readings
- Motor amps vs. rated FLA
- Seal condition/leaks
- Bearing noise or temperature
- Flow rate vs. design spec

Regulations & Codes:

NEC(NFPA 70), OSHA 1910 Subpart O, ASME B73.1 (for pumps)